

# Amir Mirzai Golpayegani

📍 Calgary, AB, Canada

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## Summary

M.Sc. Computer Science graduate with experience building Python/C++ systems for machine learning, large-scale data analysis, statistical evaluation, and data infrastructure. Skilled in PyTorch, Python data workflows, SQL/PostgreSQL, model evaluation, experimentation, technical communication, and turning complex datasets into defensible conclusions for product and research decisions.

## Education

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| <b>M.Sc. in Computer Science</b><br><i>University of Calgary</i>               | GPA: <b>3.94/4.0</b> | Sep 2022 – Feb 2026<br><i>Calgary, Canada</i> |
| <b>B.Sc. in Computer Engineering</b><br><i>Sharif University of Technology</i> | GPA: <b>3.69/4.0</b> | Sep 2016 – Feb 2021<br><i>Tehran, Iran</i>    |

## Technical Skills

**Programming:** Python, C++, JavaScript, SQL

**ML / AI:** PyTorch, Scikit-learn, TensorFlow, CNNs, statistical evaluation, LLM workflows, AI agents

**Data Science:** Experiment design, data cleaning, feature engineering, model validation, performance analysis

**Data / Geospatial:** NumPy, Pandas, SciPy, Google Earth Engine API, Discrete Global Grid System (DGGS), GDAL, Rasterio, SNAP

**Backend / Engineering:** Django, Node.js, REST APIs, PostgreSQL, Docker, Git, Linux, AWS, GitHub Actions

**Visualization / Automation:** Streamlit, Polyscope, OpenGL, GLSL, n8n, OpenClaw, SearXNG, Telegram API

## Work Experience

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| <b>Research Software Developer</b><br><i>BigGeo / Mitacs, University of Calgary</i> | Sep 2022 – Feb 2026<br><i>Calgary, Canada</i> |
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- Built C++/Python software for scalable processing, storage, validation, and retrieval of Earth observation data.
- Automated SAR and optical imagery preprocessing using GDAL, Rasterio, SNAP, and satellite data APIs.
- Developed DGGS-based encoding pipelines to convert satellite imagery into structured tensor files for analysis.
- Improved encoding performance with C++ multithreading, reducing 10-tensor processing from 233s to 26s.
- Reduced storage by 38% for scaled Canada-wide data with 1,112 GeoTIFF tiles and 16K+ DGGS tensors.

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| <b>Back-End Developer</b><br><i>Asr Gooyesh Pardaz</i> | Jun 2020 – Jul 2022<br><i>Tehran, Iran</i> |
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- Developed Django REST APIs for NLP, text-to-speech, and speech-to-text service plans backed by PostgreSQL.
- Built authentication features including registration, Google login, password reset, and email/phone verification.
- Implemented backend logic for payment-service integration and storing plan/payment records in a relational database.
- Contributed to a WebRTC-based chatbot system using GPT-style responses and FAQ data to answer client questions.

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| <b>Deep Learning Teaching Assistant</b><br><i>Neuromatch Academy</i> | Jul 2026<br><i>Remote, United States</i> |
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- Mentored an international pod of ~15 students through Neuromatch Academy's intensive three-week Deep Learning course.
- Facilitated hands-on PyTorch tutorials spanning MLPs, CNNs, attention/transformers, diffusion generative models, and reinforcement learning.
- Guided project groups through end-to-end deep learning research projects, from question formulation and literature review to model implementation and final presentations.

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| <b>Teaching Assistant</b><br><i>Department of Computer Science, University of Calgary</i> | Dec 2022 – Apr 2025<br><i>Calgary, Canada</i> |
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- **Developing Big Data Applications:** Mentored +25 students on AWS-based data pipelines, cloud storage, debugging, and scalable data workflows.
- **Working with Data at Scale:** Supported +20 students with large-scale data processing, data pipelines, and high-volume data analysis.
- **Database Management Systems:** Assisted +75 students with SQL, database design, course projects, grading, and exam support.

## Research and Technical Projects

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| <b>Spatiotemporal Earth Observation Data System Using DGGS</b><br><i>First-author publication   M.Sc. thesis   C++, Python, DGGS, Wavelets, B-splines, Time-Series</i> | Sep 2022 – Feb 2026 |
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- Designed a DGGS-based framework for real-time multiresolution management of satellite imagery.

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| <ul style="list-style-type: none"> <li>Developed a tensor-based storage model for direct access to arbitrary regions at multiple resolutions.</li> <li>Implemented a triangular wavelet scheme to support multiresolution retrieval without image-pyramid redundancy.</li> <li>Built B-spline temporal approximation to compress time-series data and retrieve missing timestamps locally.</li> <li>Retrieved 105,489 DGGs cells per time-series query in 13.9s, compared with about 220s using global retrieval.</li> </ul>                                                                                                          |                     |
| <b>DEM Super-Resolution Framework</b><br><i>Python, PyTorch, CNNs, Scientific Imaging, Google Earth Engine API, Polyscope, Data Processing and Visualization</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | May 2024 – Feb 2026 |
| <ul style="list-style-type: none"> <li>Trained deep ResNet models for 4× DEM super-resolution, reconstructing 30m elevation data from 120m inputs.</li> <li>Built a Streamlit pipeline using Rasterio to parallelize GEE DEM tile downloads and systematically collect and preprocess mountainous terrain data worldwide.</li> <li>Implemented PyTorch training and evaluation workflows with terrain-aware metrics such as RMSE, slope, TRI, TPI, and wavelet-based errors.</li> <li>Developed a Polyscope-based 3D visualization tool to render DEM tiles from multiple angles and compare reconstruction quality in 3D.</li> </ul> |                     |
| <b>Agentic Job Search Automation Platform</b><br><i>OpenClaw, n8n, LLMs, AI Agents, Docker, Workflow Automation, SearXNG, Telegram API</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | May 2026 – Present  |
| <ul style="list-style-type: none"> <li>Building an agentic AI system to search jobs, analyze job fit, and generate tailored application materials.</li> <li>Designed LLM-based workflows for resume/job matching, skill-gap analysis, and application decision support.</li> <li>Integrated Google Sheets, Google Drive, and Telegram API to track jobs and send automated application alerts.</li> <li>Prototyped the initial workflow in n8n before migrating toward a more flexible OpenClaw-based agent system.</li> </ul>                                                                                                        |                     |
| <b>Persian Text Normalizer for NLP</b><br><i>Python, NLP, Text Processing</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Oct 2016 – Feb 2021 |
| <ul style="list-style-type: none"> <li>Developed a Python-based text normalization toolkit to standardize Persian text for downstream NLP pipelines.</li> <li>Implemented preprocessing rules for cleaning, formatting, and preparing Persian text data for language-processing tasks.</li> </ul>                                                                                                                                                                                                                                                                                                                                     |                     |
| <b>Publication</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                     |
| <b>Amir Mirzai Golpayegani</b> , Mahmudul Hasan, Faramarz F. Samavati. <i>Real-Time Multiresolution Management of Spatiotemporal Earth Observation Data Using DGGs</i> . Remote Sensing, 2025. <a href="https://doi.org/10.3390/rs17040570">doi:10.3390/rs17040570</a>                                                                                                                                                                                                                                                                                                                                                                |                     |
| <b>Certificates</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                     |
| <b>Supervised Machine Learning: Regression and Classification</b> ,<br>DeepLearning.AI and Stanford University, Coursera                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Oct 2024            |
| <b>Applied Machine Learning in Python</b> ,<br>University of Michigan, Coursera                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Jul 2020            |
| <b>Languages</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                     |
| <b>English:</b> Full Professional Proficiency (TOEFL 115/120)<br><b>Persian:</b> Native Proficiency                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                     |
| <b>Achievement</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                     |
| <b>Teaching Assistant Excellence Laureate</b><br>Department of Computer Science, University of Calgary — awarded for outstanding teaching support ( <a href="#">link</a> )                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 2024/2025           |
| <b>Ranked 17th Nationwide in Iran University Entrance Exam (Konkour)</b><br>Top 0.001% among all participants                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | May 2016            |